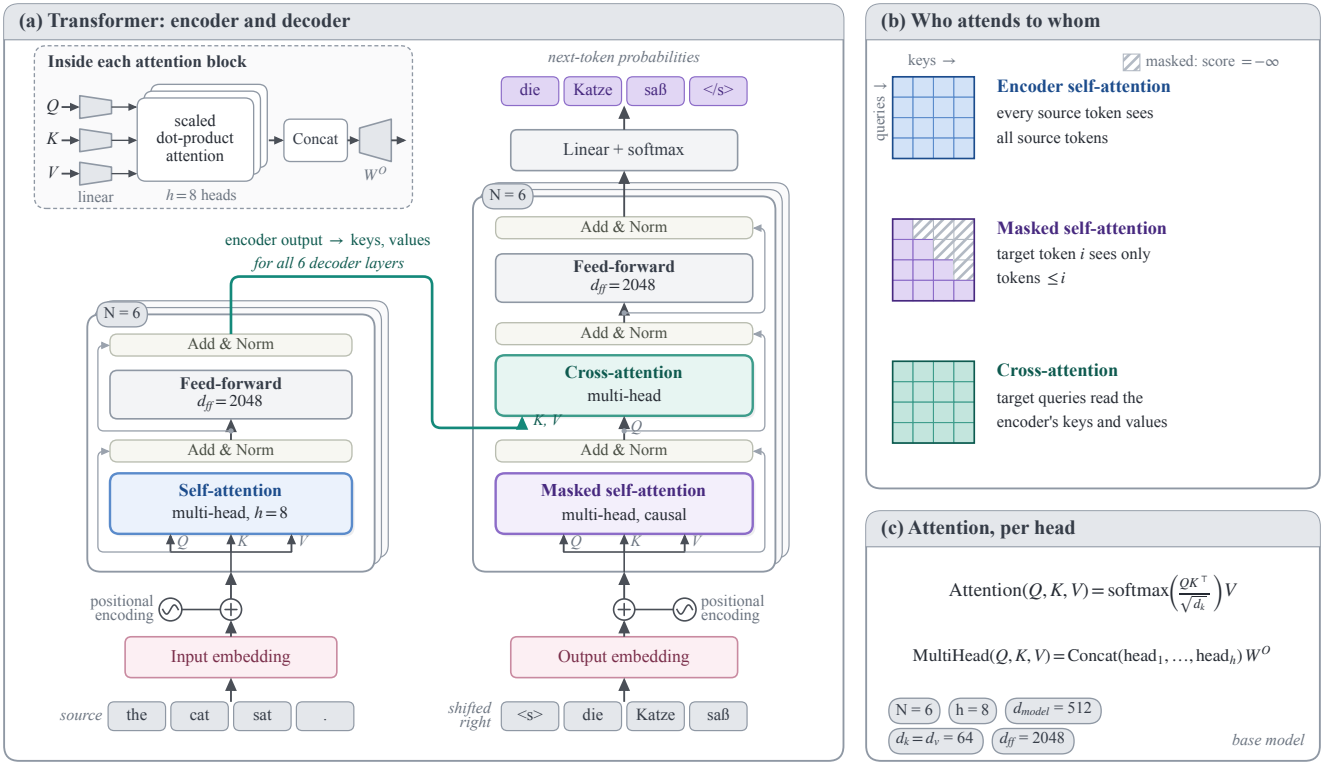

(a) Transformer: encoder and decoder
Inside each attention block
Q
K
V
linear
scaled dot-product attention
h = 8 heads
Concat
W^O
next-token probabilities
die
Katze
saß
</s>
Linear + softmax
N = 6
Add & Norm
Feed-forward
d_ff = 2048
Add & Norm
Cross-attention
multi-head
K, V
Q
Add & Norm
Masked self-attention
multi-head, causal
Q K V
encoder output → keys, values
for all 6 decoder layers
N = 6
Add & Norm
Feed-forward
d_ff = 2048
Add & Norm
Self-attention
multi-head, h = 8
Q K V
positional encoding
Input embedding
source
the
cat
sat
.
positional encoding
Output embedding
shifted right
<s>
die
Katze
saß
(b) Who attends to whom
keys →
queries ↓
masked: score = −∞
Encoder self-attention
every source token sees all source tokens
Masked self-attention
target token i sees only tokens ≤ i
Cross-attention
target queries read the encoder's keys and values
(c) Attention, per head
Attention(Q, K, V) = softmax(QK^T / √d_k) V
MultiHead(Q, K, V) = Concat(head_1, …, head_h) W^O
N = 6
h = 8
d_model = 512
d_k = d_v = 64
d_ff = 2048
base model